

Physiological Data Visualization Website

Purpose: Build an interactive web application that lets a user select a conversation context, view the conversation turn-by-turn, and simultaneously inspect five physiological signals: PR, SkinTemp, EDA, ACCEL, ACT_CLASS. Selecting a turn highlights the corresponding time interval on every physiological graph.

1. Project Overview

The website is intended to present conversational data together with physiological data collected during the conversation. The application should make it easy to understand how physiological signals changed over time and how those changes relate to individual conversation turns.

Primary user flow

The user opens the website and sees a list/grid of available conversation contexts.

The user selects one context.

The application loads the conversation turns and the associated physiological graphs.

The page displays the conversation on the left and five synchronized physiological graphs on the right.

The user clicks a conversation turn.

The application determines the time interval associated with that turn from the supplied physiological-summary JSON.

That interval is highlighted on all five graphs.

The user can click another turn to move the highlight, or return to the context-selection screen.

2. Functional Requirements

2.1 Context Selection Interface

Display all available **conversation contexts** as cards, buttons, or a clean list.

Each context should have a unique `context_id`.

Optional metadata may be displayed, such as participant ID, date, or short **scenario** description.

Clicking a context should navigate to the conversation + physiology visualization page.

The application **does not** require the developer to hard-code each context into the frontend; contexts should ideally be loaded from a metadata/configuration file.

2.2 Conversation View

Display the conversation as sequential turns: Turn 1, Turn 2, Turn 3, etc.

Each turn should show at minimum: turn number, speaker/user ID, user message, AI Strategy, AI response, and the physiological summary.

The selected turn must have a clearly visible active/highlighted state.

Clicking a turn must update the highlighted region on all five graphs.

The conversation panel should be scrollable independently if the conversation is long.

The currently selected turn should remain visually identifiable.

2.3 Physiological Graphs

The right side of the page should contain five time-series graphs. The exact five signal names must be finalized from the supplied CSV files.

One graph (showing the trend for the entire conversation) per physiological signal.

Shared time axis so that all five graphs are temporally aligned.

X-axis: timestamp/time elapsed.

Y-axis: signal-specific value and unit.

Hover tooltip showing exact timestamp and signal value.

Zoom/pan should be optional but useful for long conversations.

The selected turn's time interval must be visually highlighted on every graph.

Graphs should remain readable on laptop/desktop screens and adapt reasonably to smaller screens.

2.4 Turn-to-Time Highlighting

This is the central interaction of the application. The physiological-summary JSON should provide, directly or indirectly, the time interval represented by each turn.

Example: if Turn 4's physiological summary says 'in the past 2 minutes', the application should identify the corresponding two-minute interval and highlight it.

The highlight should appear as a translucent vertical band spanning the complete graph height.

All five graphs must highlight exactly the same time range.

Changing the selected turn should remove the previous highlight and apply the new interval.

If the summary contains an explicit start/end timestamp, use those timestamps rather than trying to infer the interval from text.

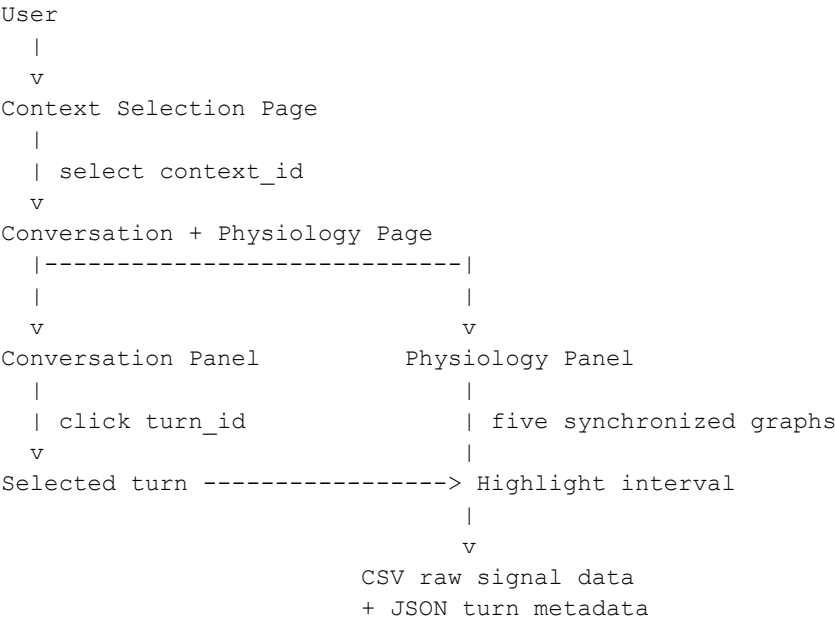
2.5 Data Loading

Raw physiological values will be provided as **CSV files** for each participant.

Physiological summaries and conversation/turn metadata will be provided as **JSON files**.

For the first version, local/static JSON and CSV files can be loaded from the application's public/data directory.

3. High-Level Application Architecture



4. Suggested Page Layout

Context: Academic Stress		[Back]
Conversation / Turns	Physiological Signals	
Turn 1	EDA graph	
User: ...	[highlighted interval]	
Assistant: ...		
	PR graph	
Turn 2 <-- selected	[highlighted interval]	
User: ...		
Assistant: ...	Skin Temperature graph	
	[highlighted interval]	
Turn 3		
...	ACCEL graph	
	[highlighted interval]	
	ACT_CLASS graph	
	[highlighted interval]	

A two-column layout is recommended: approximately 35–40% width for the conversation and 60–65% for the graphs. The graphs should remain vertically aligned and use a shared visual time range.